



MECHANIC TO PENTESTER

Hypothesis-Based Testing Across
Industries

DIAGNOSTIC REASONING

The process of inferring an unobservable condition based on indirect observations from diagnostic tests.

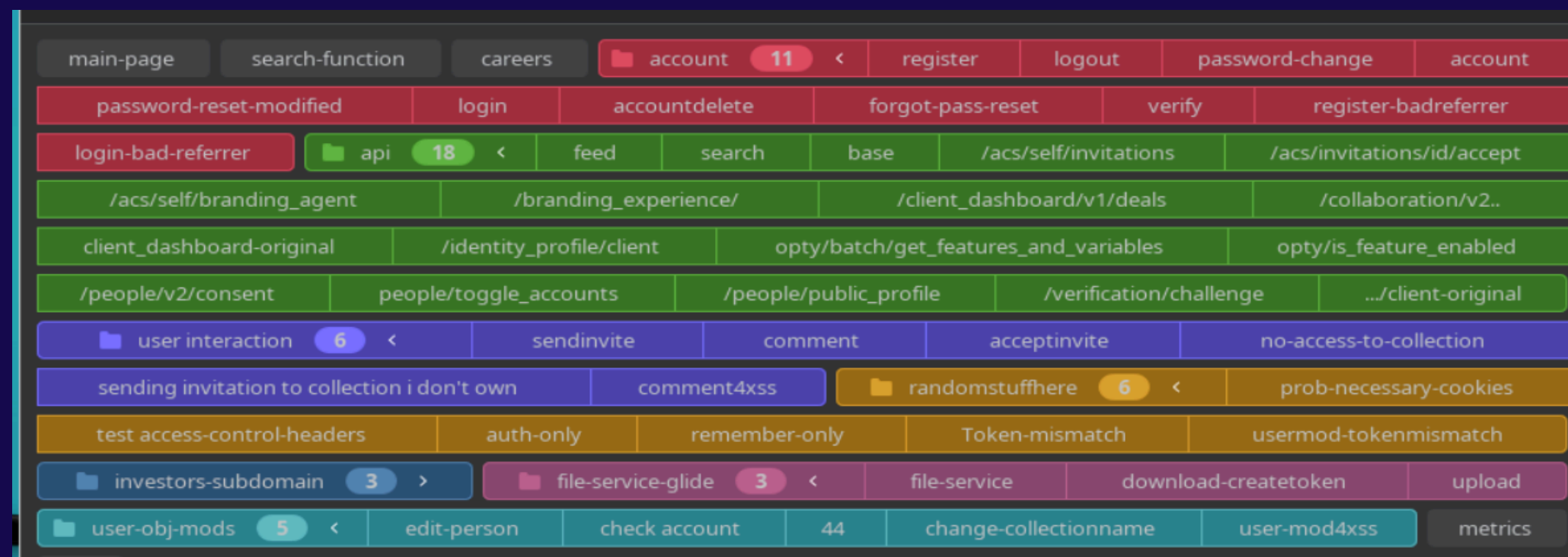


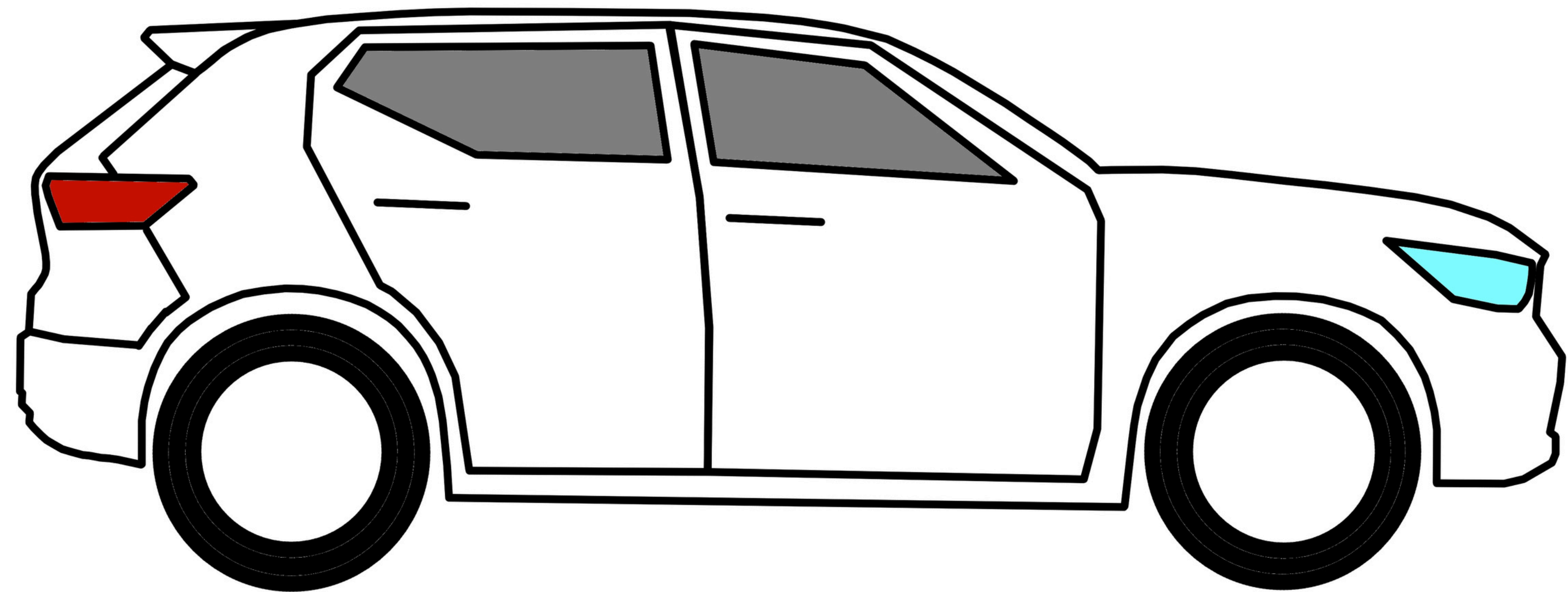
Application Security Specialist
@ Digital Boundary Group

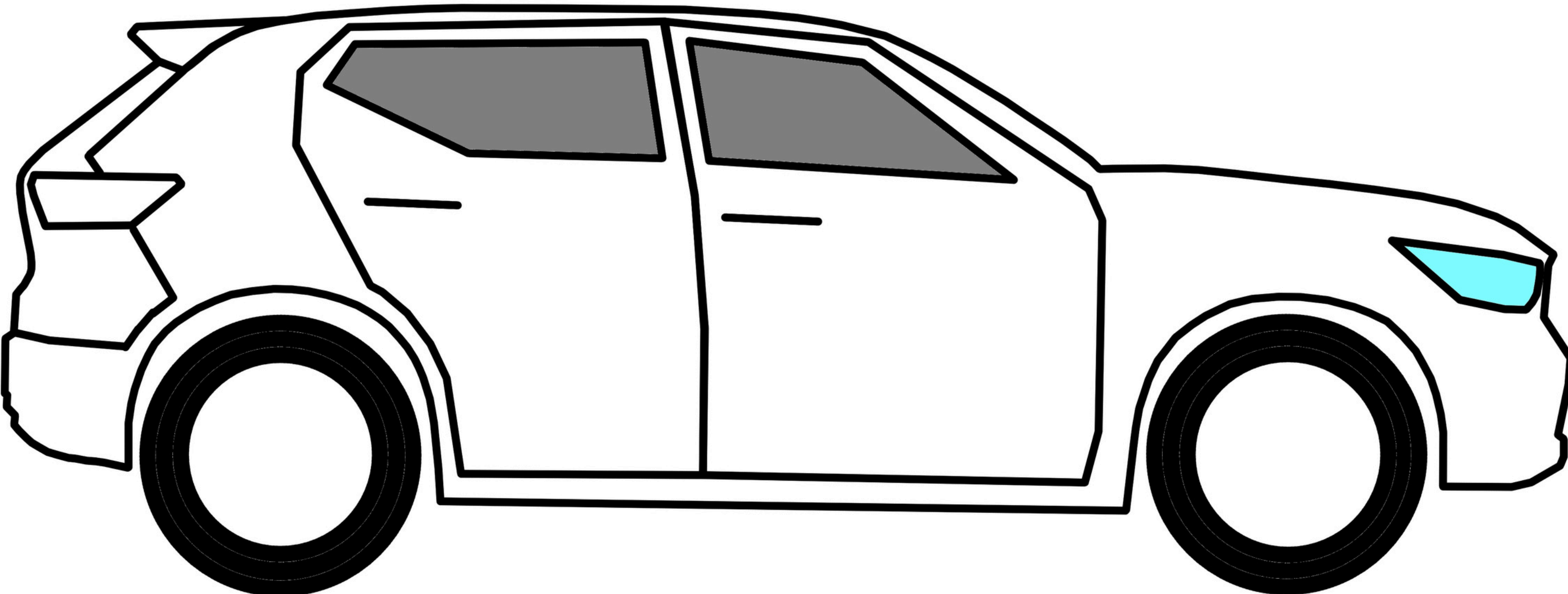
Security Researcher

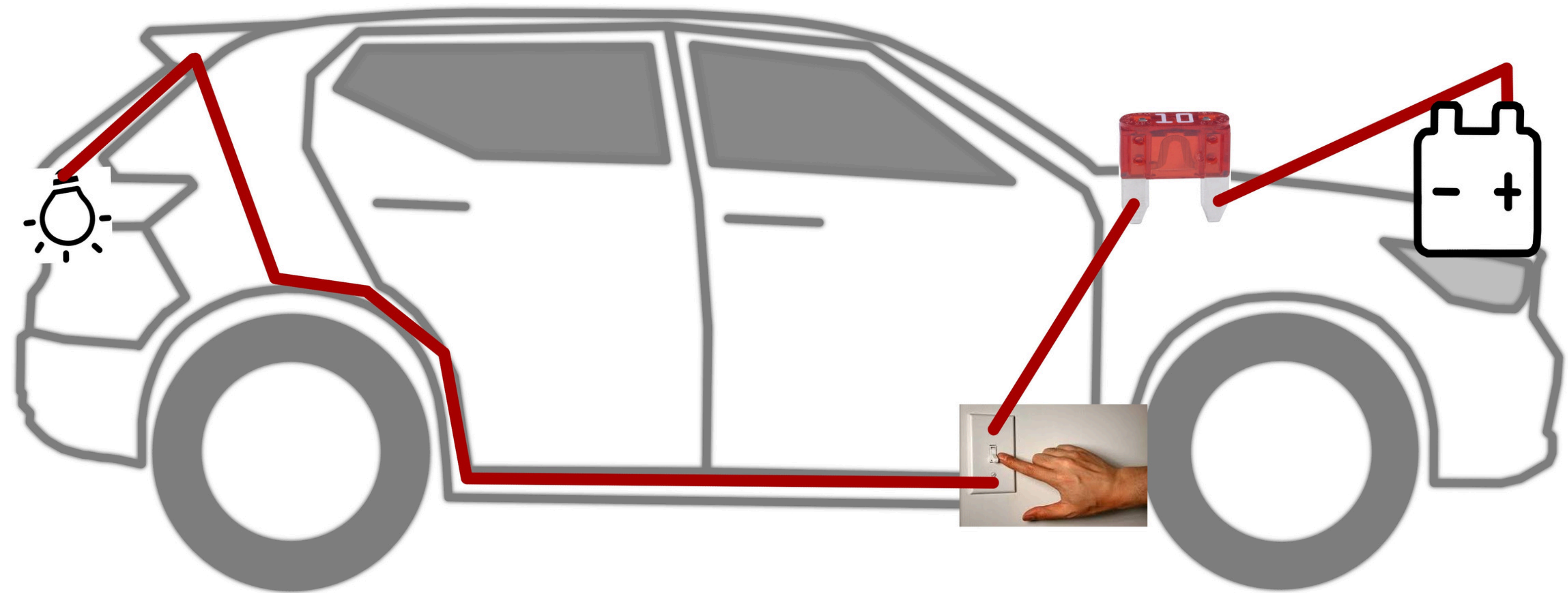
Bug Bounty Hunter

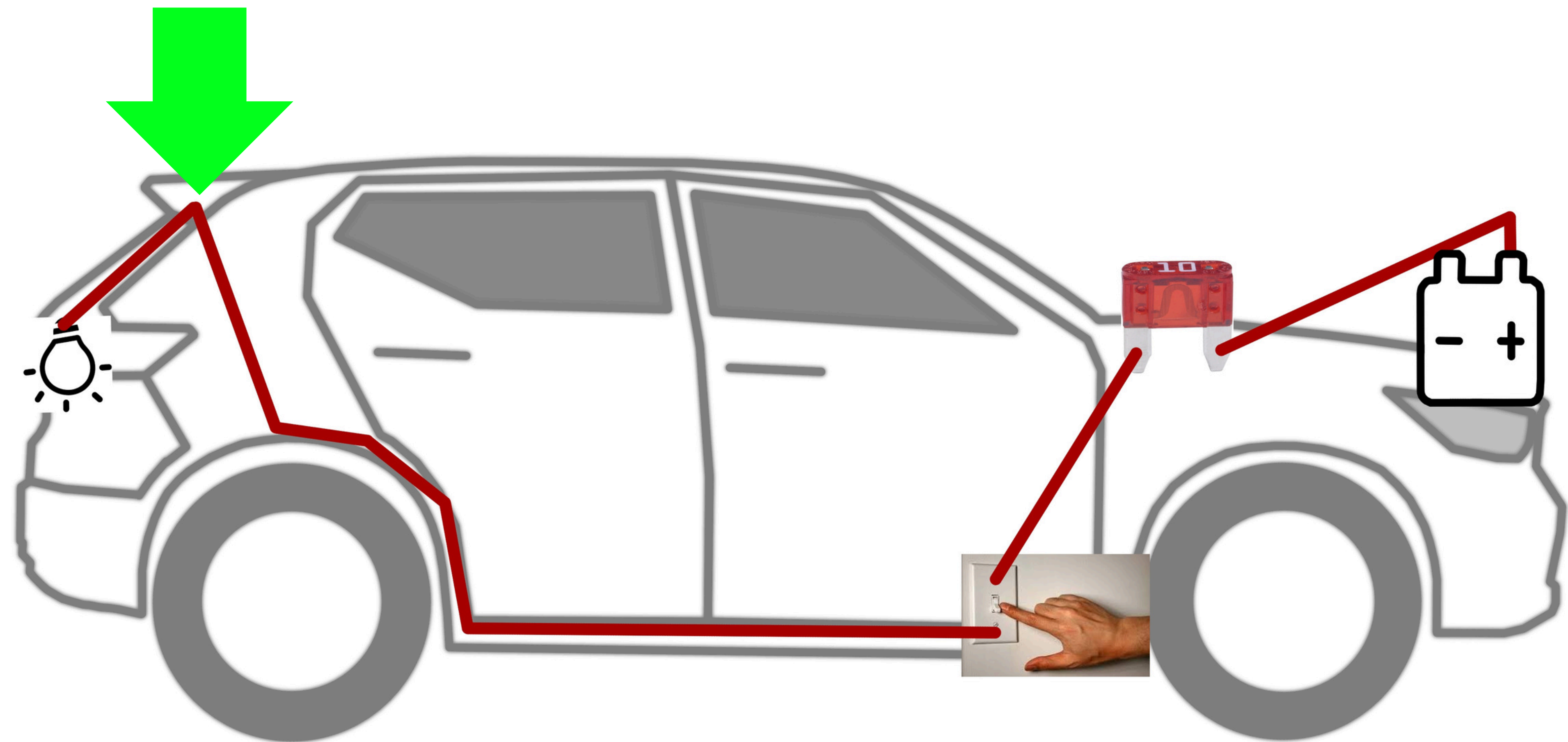
WHOAMI

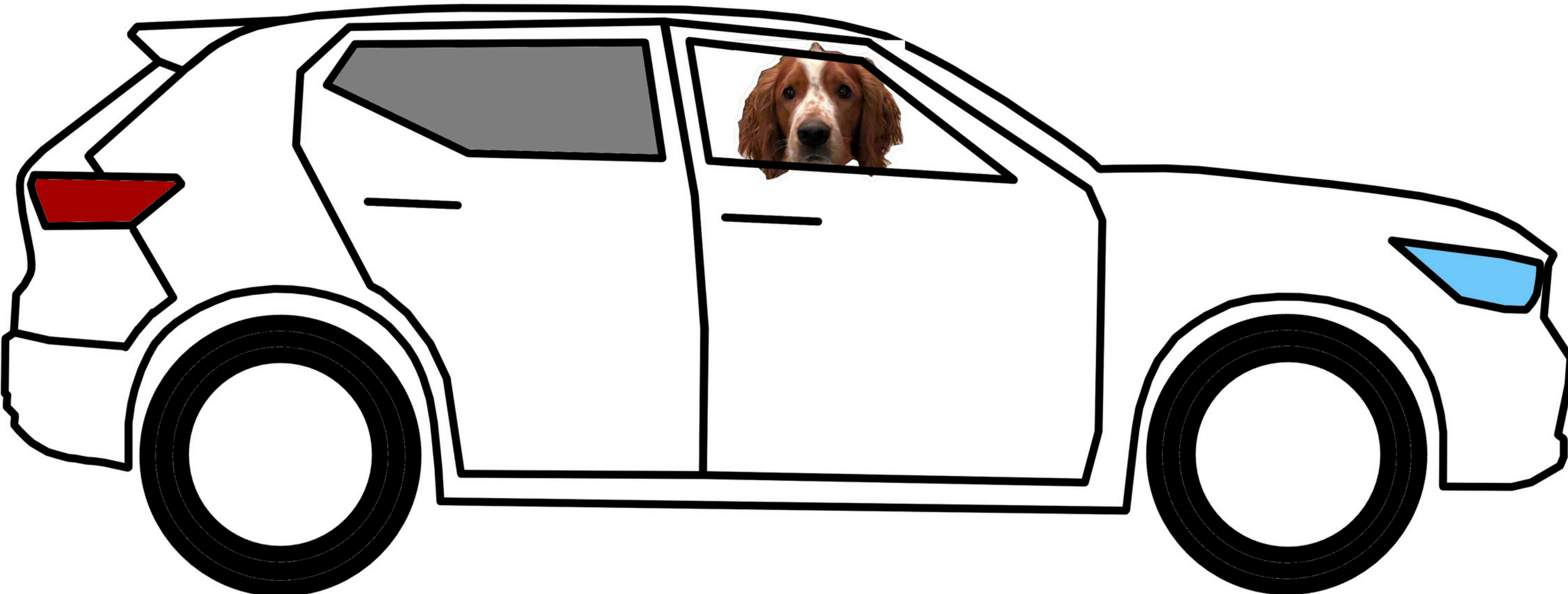


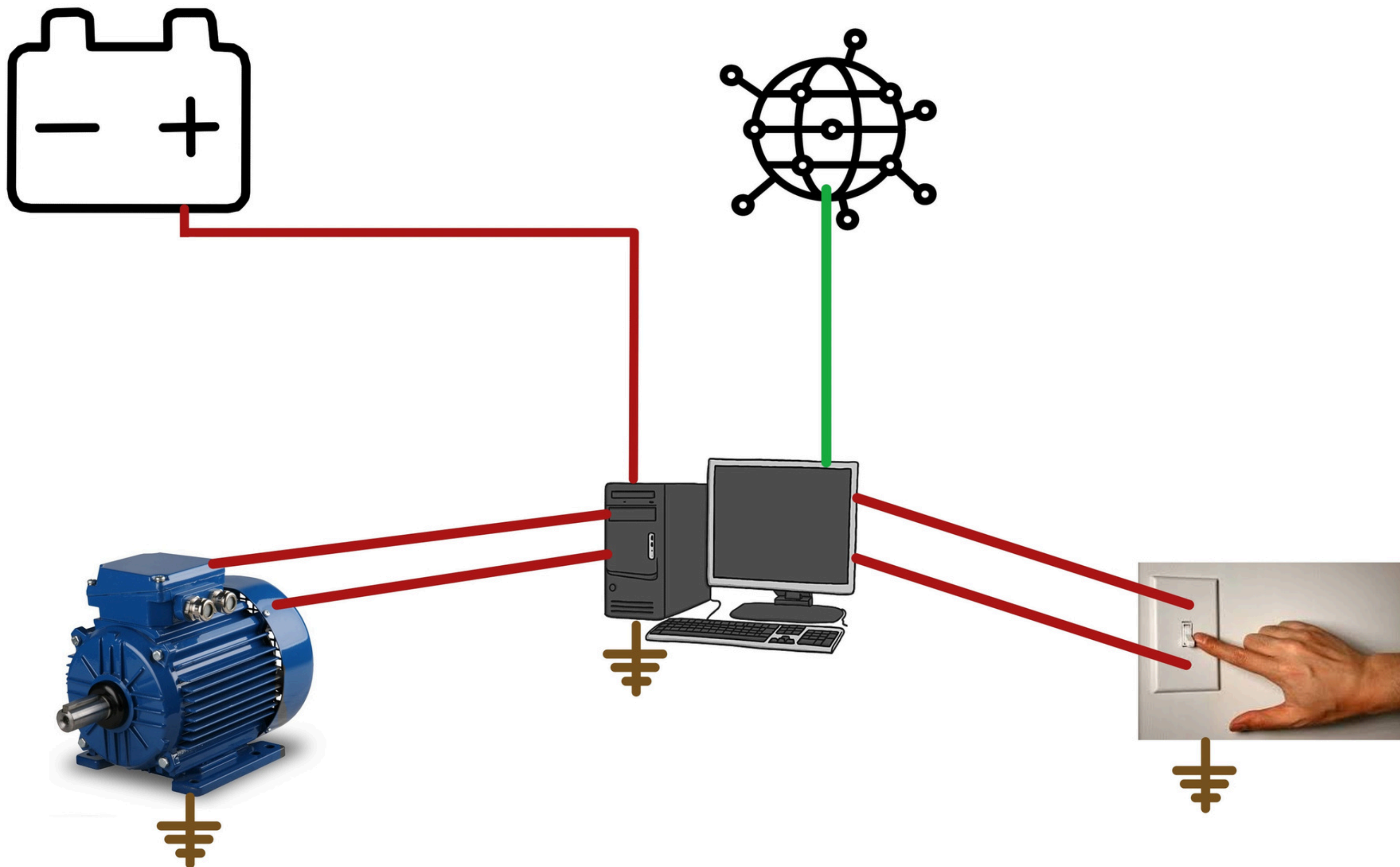


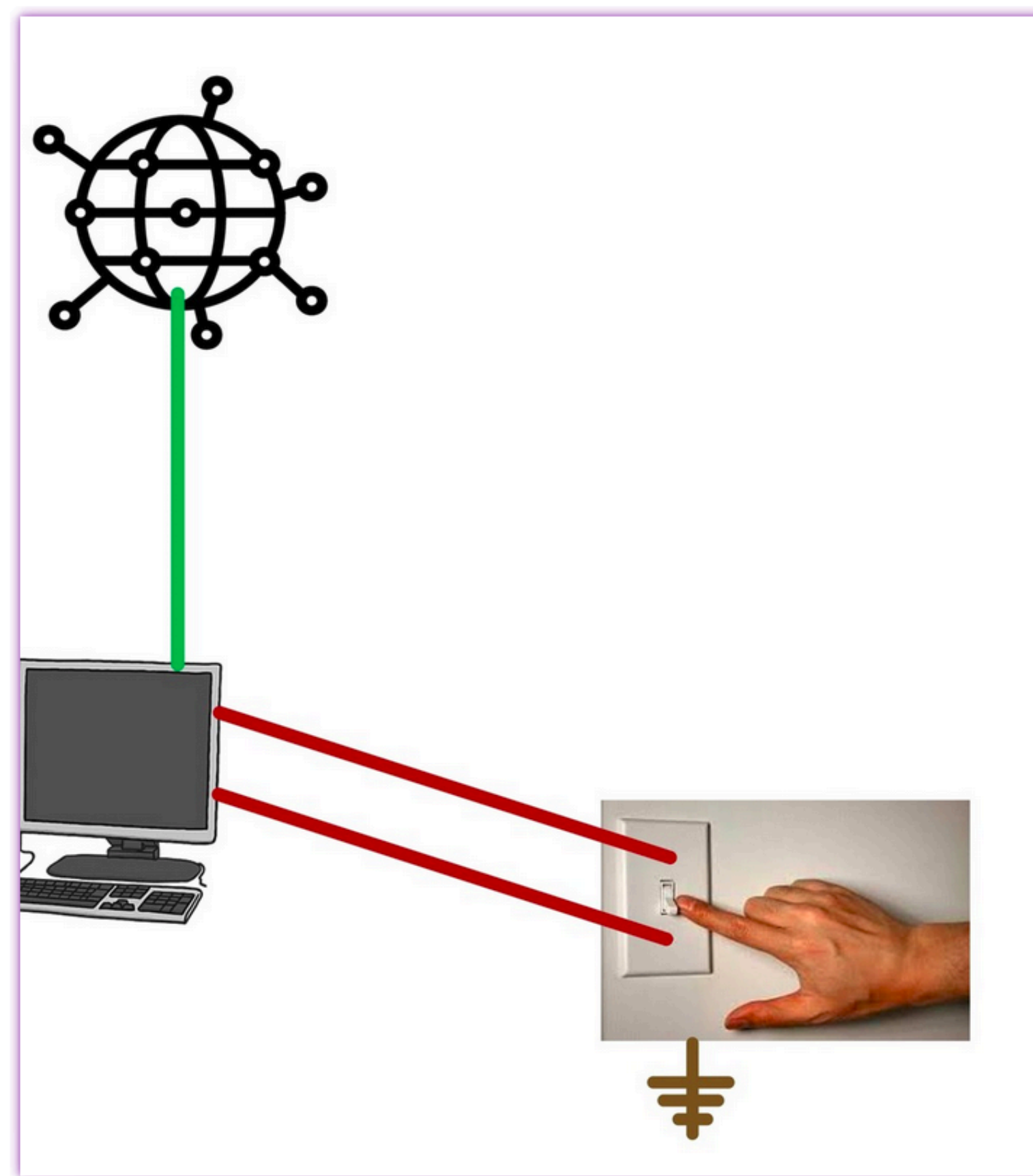
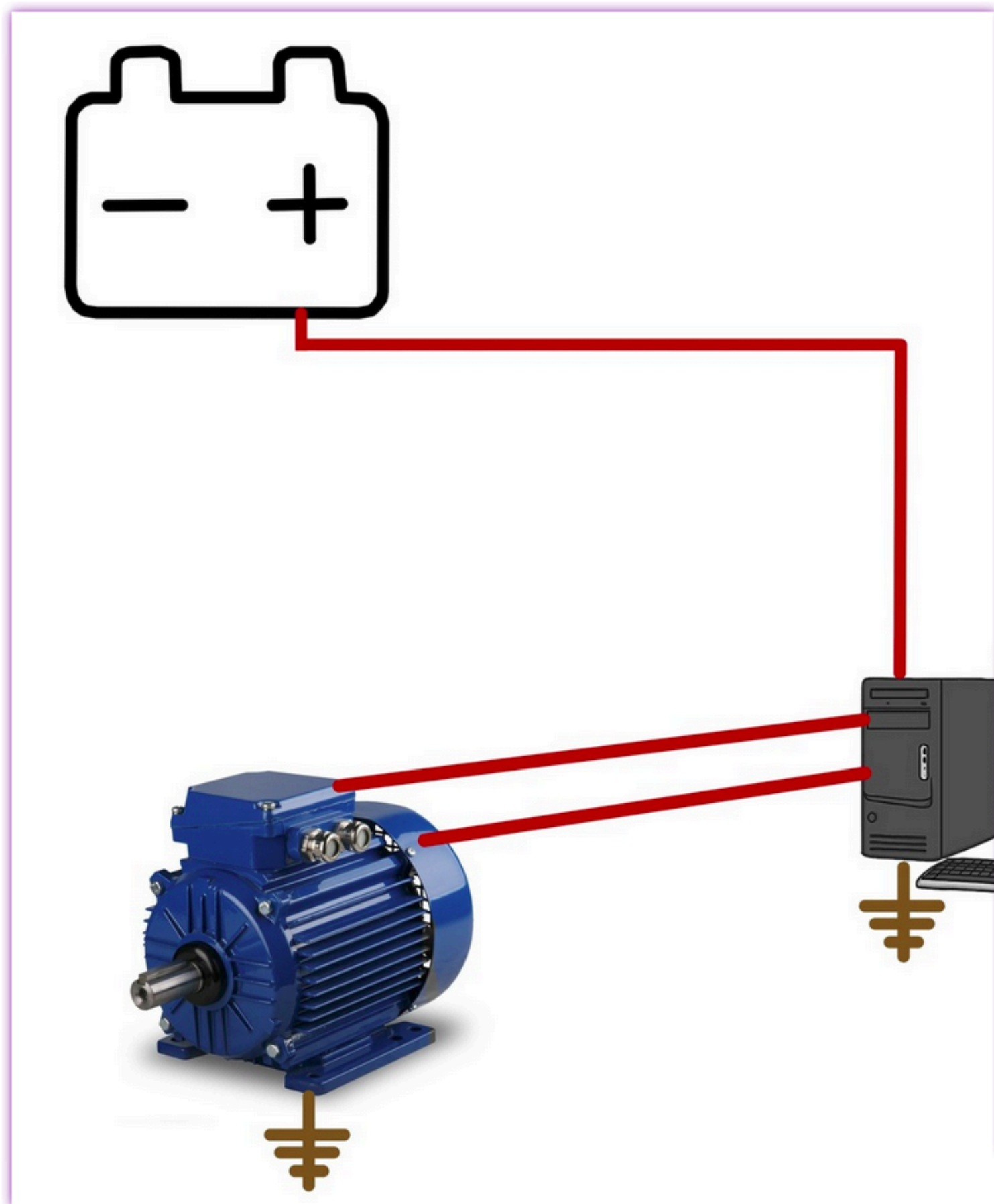






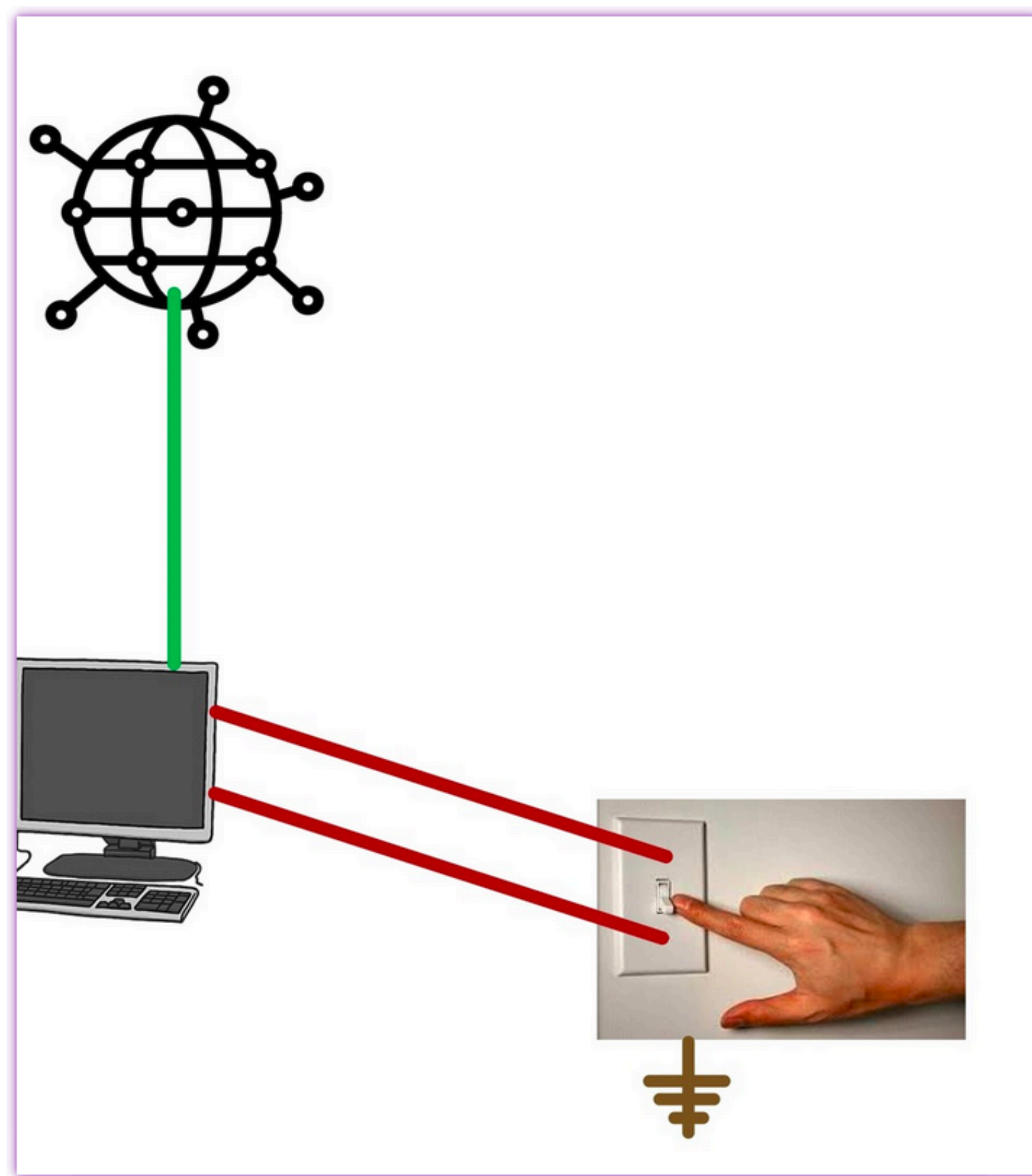
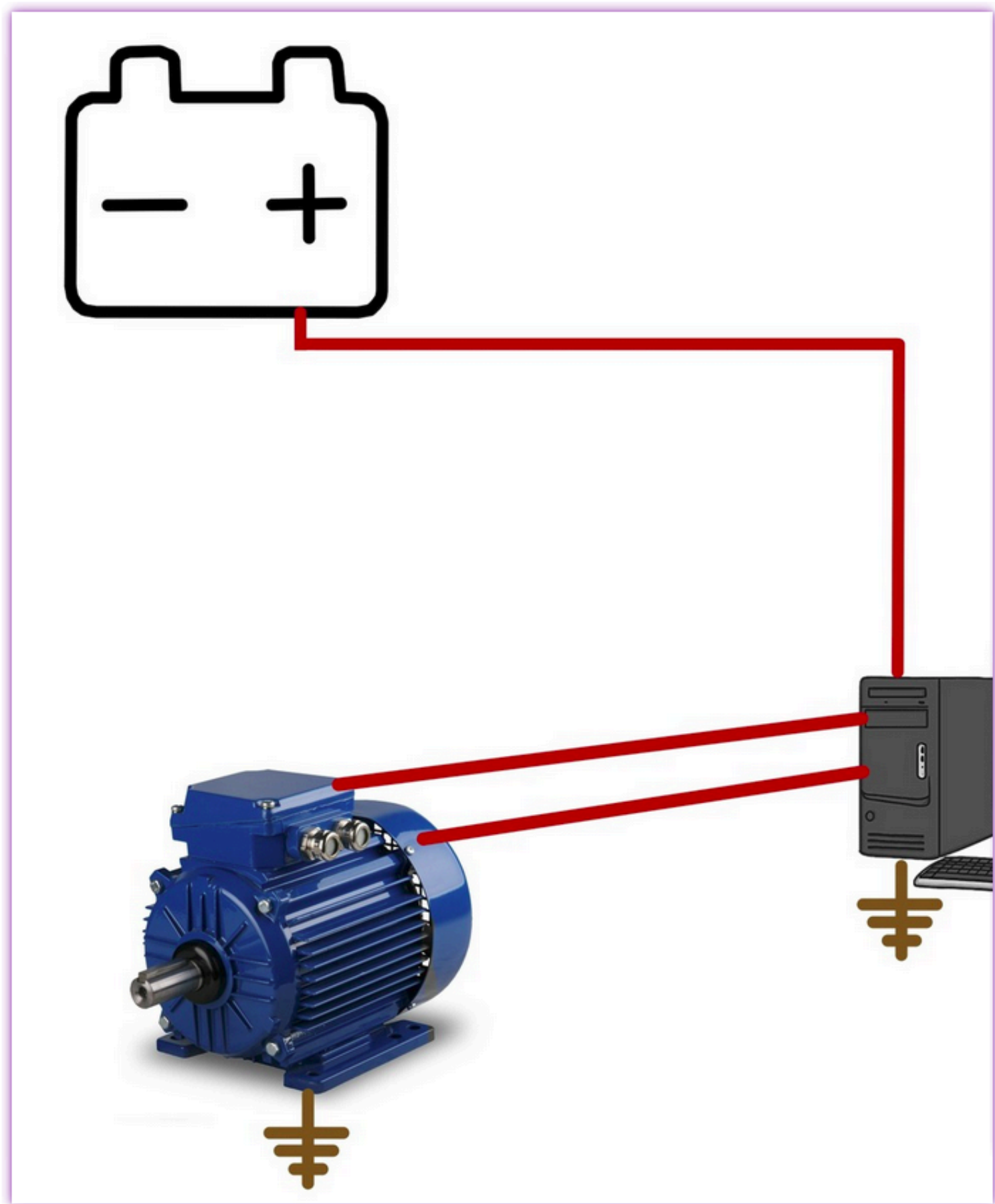






- Battery could be bad
- Battery could be discharged
- Battery could be missing
- Battery could be disconnected
- Battery fuse could be blown
- Battery power connection could be loose
- Battery ground connection could be loose
- Battery ground cable could be severed
- Battery ground cable could be corroded
- Battery to fuse box wire could be open
- Battery to fuse box wire could be corroded
- Alternator could be bad
- Serpentine belt could be missing
- Voltage regulator could be bad
- Fuse could be blown
- Fuse could be missing
- Incorrect fuse installed
- Fuse box could be broken
- Fuse box connections could be loose
- Wires off fuse box could be corroded
- Wires off fuse box could be open
- Power wire through door pivot could be severed
- Network wires through door pivot could be severed
- Short to power in harness at door pivot
- Insulation damaged on power wire in harness at door pivot
- Insulation damaged on network wires in harness at door pivot
- Door hinge and hardware loose
- Corrosion in wires at door pivot
- Corrosion in wires under floor
- Harness unplugged at motor
- Motor ground wire open
- Motor ground wire short to power
- Motor ground wire corroded
- Motor ground wire partially damaged
- Motor power wire open

- Motor power wire short to ground
- Motor signal wire short to ground
- Motor open
- Motor internal short
- Motor short to case (ground)
- Motor physical damage
- Motor stuck
- Low quality motor
- Harness unplugged at switch
- Switch open
- Switch internal short
- Switch Signal 1 short to ground
- Switch Signal 2 short to ground
- Switch corroded connections
- Switch physical damage
- Low quality switch
- Door connector could be open
- Door connector could be corroded
- Door connector position assurance lock could be broken
- Door connector pin could be bent
- Door connector pin could be missing
- Door connector wires could be lose
- Wires from door connector to body electrical could be open
- Wires from door connector to body electrical could be corroded
- Wires from door connector to body electrical could be short to ground
- Wires in door hinge could be open
- Wires in door hinge could be damaged
- Wires in door hinge could be corroded
- Wires in door hinge could be short to ground
- Wires in door hinge could be short to power
- Wires in door hinge could have intermittent short.
- Door harness could be damaged from poor repair attempt
- Door harness could be damaged from incorrect previous installation
- Door harness could have swapped wires from previous repair
- Switch connector could have swapped wires from previous repair.



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- Door harness could have swapped wires from previous repair
- Switch connector could have swapped wires from previous repair.

- Motor connector could have swapped wires from previous repair.
- Computer connector could have swapped wires from previous repair.
- Computer could have high resistance to ground.
- Computer could have no power.
- Computer could be in safe mode due to incorrect signal from another computer.
- Computer could be off and not turning on.
- Computer could have no ground.
- Computer could have burnt components.
- Computer could have water damage.
- Computer could have physical damage.
- Computer could require software update.
- Technical bulletin for additional parts installation for durability of system.
- Technical bulletin for physical repair.
- Technical bulletin for inspection.
- Technical bulletin for software.



If I run commandline recon tools I will learn the most likely place to find a vulnerability!

```
—<sitemap>
```

```
—<loc>
```

```
https://www.██████████.com/content/██████████/  
en.sitemap.campaigns-sitemap.xml
```

```
</loc>
```

```
<lastmod>2026-01-01██████████</lastmod>
```

```
</sitemap>
```

```
—<sitemap>
```


FORM

Title *

DEMO

Details * PREVIEW

B *I* U ~~S~~ I_x 😊 🔗 📷 🎥  ▼  ▼ HTML 🗑️

🔗 **A** ▼ **T**! ▼ A ▼ `</>`      

`<image src=1>`

Hint: @ links to members

Your content was last auto-saved at 12:31 AM

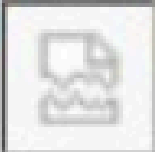
Topic tags 7 characters remaining

`<image src=1>`

`<img src=0 onerror=alert(document.domain)>`

EXECUTION

Topics with Tag: `<image src=1>`

1 Results for "**"**

FORM

Title *

DEMO

Details * PREVIEW

B *I* U ~~S~~ I_x 😊 🔗 📷 🎥  ▼  ▼ HTML 🗑️

🔗 **A** ▼ **T**! ▼ A ▼ `</>`    

`<image src=1>`

Hint: @ links to members

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A Reachable Assertion

1. Service worker installation by the parent thread
2. Loaded directly into memory
3. Parsed with string functions
4. Hit an assertion error

Credit for CVE-2026-0889 shared with Elysee Franchuk

Hypothesis

Is there no resource consumption limits for Service Worker import except for this assertion error?

Memory Exhaustion Attack

192.168.250.1/memory.htm

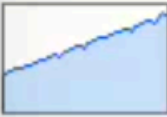
Memory Exhaustion Demo

This page attempts to exhaust the host machines resources by registering multiple service workers with the largest individual size accepted by firefox.

Performance



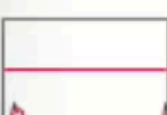
CPU
100% 2.61 GHz



Memory
14.7/16.0 GB (92%)



Disk 0 (C:)
SSD (NVMe)
11%



Ethernet
Ethernet0
S: 0.1 R: 19.4 Mbps

Memory

Memory usage



60 seconds

Memory composition

In use (Compressed)

14.6 GB (68.4 MB)

Available

1.3 GB

Slots

Hard

Committed

15.6/17.0 GB

Cached

1.1 GB

Paged pool

135 MB

Non-paged pool

92.3 MB

TEACHING CRITICAL THINKING

PRINCIPLES

- Manage **MULTIPLE** hypotheses at the same time
- **NEVER** stop looking for new hypotheses
- Prioritize the **RATE** of information gain
- **EVALUATE** your test results
- **CANNOT** diagnose through elimination